

Referee Feedback: Summary and Revision Notes

MS-MKG-2026-01033 — “Consumer Behavior and Algorithmic Price Competition: The Case of Reference Dependence”

Internal discussion document • Decision: **Reject** (Management Science)

At a glance

Editor: Greg Shaffer (DE). Three referee reports + an Associate Editor report; AE recommended reject, DE concurred.

The single criticism that drove the decision (Reviewers 2 and 3, endorsed by the AE):

Under rational expectations the reference price equals the actual price on the equilibrium path ($r = p$), so the reference term $\gamma(r - p) = 0$ there. Hence Proposition 1 reduces to a *standard price-elasticity comparative static* rather than a distinct reference-dependence result. Rewriting Eq. (1) as $u = \theta - (1 + \gamma)p + \gamma r + \varepsilon$ makes γ look like it is simply rescaling price sensitivity.

Everything else (theory–simulation misalignment, “open the black box,” positioning) is serious but secondary to this identification concern. **Our revision must first establish that reference dependence does something *beyond* elasticity rescaling** — theoretically (off the equilibrium path) and in the simulations.

1 Decision letter (Shaffer)

- Topic is important and timely; contribution judged insufficient for *Management Science*.
- Primary concern: **lack of alignment between the theoretical model and the simulation framework**.
- Secondary: concerns about the simulations themselves, and whether the theoretical results are an **artifact of how reference dependence is modeled**.

2 Associate Editor report

AE1. Theory–simulation alignment (all reviewers). Theory is a static duopoly (Nash); simulations are dynamic algorithmic learning. R2/R3: the paper essentially studies *increased price sensitivity*, not reference dependence. R3: under rational expectations the main result is a standard elasticity comparative static. $\Rightarrow$ *a dynamic model is needed*.

AE2. Concerns about the simulations. R1: without human subjects, can simulations establish the implications of consumer biases? (Noted as *not* a major limitation *if* the theory were strong.) R2: multi-agent learning $\Rightarrow$ multiple equilibria; uniqueness not guaranteed. R3: simulations alone cannot establish that reference dependence curbs collusion, and do not reveal the driving mechanism.

AE3. Overall. Substantially strengthen the theory, integrate theory and simulation more closely, and provide stronger evidence for the mechanisms. “Rethink both the model and the analyses.”

3 Reviewer 1

- R1.1. Theory–simulation misalignment.** Simulations use n firms in a Markov decision process; theory is a duopoly with static Nash. Extend the theory to multinomial logit with n firms and characterize a **Markov perfect equilibrium** via dynamic programming.
- R1.2. Single γ ignores loss-aversion asymmetry.** Prospect theory: losses loom larger than gains, so effects should differ above vs. below the reference. The “prices are usually below reference” defense assumes consumers don’t communicate and firms fully set references.
- R1.3. AI motivation not modeled.** AI motivates the paper but nothing in the theory is AI-specific; algorithmic pricing is not unique to AI. If AI is tangential, other channels (e.g. social media shaping reference prices) are equally relevant. Align motivation and analysis.
- R1.4. Exogenous reference price.** The disciplining result depends on references being outside firms’ control. In practice firms manipulate references (e.g. MSRP). Endogenous reference-setting could qualitatively reverse the results.
- R1.5. Simulations cannot capture human biases.** Recommends lab experiments with human subjects to validate the predictions.

4 Reviewer 2

- R2.1. γ confounds price sensitivity and reference dependence (the “fundamentally critical” point).** $u = \theta - (1 + \gamma)p + \gamma r + \varepsilon$: any linear utility cannot isolate a distinct reference effect. If γ is just price sensitivity, the paper reduces to “more price-sensitive consumers $\Rightarrow$ lower prices, profits, collusion.” **Proposed test:** run a benchmark with $u = \theta - \beta p$ (no reference term) and show results differ qualitatively — though they are skeptical they will.
- R2.2. Proposition 2 overclaims.** Appendix A.1.4 establishes the sign of $d\bar{\delta}/d\gamma$ only *locally at* $\gamma = 0$ and in *limiting regimes* ($\mu \rightarrow \infty$, $c \rightarrow \theta$, $\mu \rightarrow 0$). It does not prove the global claim that for “large enough” μ or c collusion gets easier over the whole $\gamma \in [0, 3]$ range.
- R2.3. Q-learning convergence overclaiming & equilibrium selection.** The environment is a multi-agent Markov game with an evolving reference process — not a stationary single-agent MDP — so Watkins–Dayan guarantees do not apply. Equilibrium multiplicity means a selection mechanism operates silently. Rewrite the convergence discussion; be transparent about selection.
- R2.4. Positioning.** The introduction reads as an AI-consumer-agent paper, but $\sim 90\%$ is about reference dependence; AI agents appear only in Section 7, and it is unclear that reference prices matter for the AI-agent interplay.

5 Reviewer 3

- R3.1. What does reference dependence add beyond elasticity rescaling?** With rational expectations $r = p$ on path, so $\gamma(r - p) = 0$ and Prop. 1 is the standard “more elastic demand $\Rightarrow$ lower markups” result (consistent with Köszegi–Rabin: gain–loss utility is muted when outcomes are anticipated). The *simulations* have cycles where $r \neq p$, so the simulated decline may come from realized reference gaps — a *different* mechanism than the theory. Need a model where r and p can differ on the equilibrium path (uncertainty, dynamic reference formation).

- R3.2. Claims exceed the evidence.** “Reference dependence is a competitive safeguard” rests on one specific Q-learning environment with many modeling choices. Provide stronger theory or real-world validation; otherwise narrow the claim (which then may not clear the MS bar).
- R3.3. Open the black box.** The paper reports outcomes but not learned policies. Are algorithms colluding less? Different punishment strategies? Weaker deviation incentives? Do Calvano-et-al.-(2020)-style diagnostics: best-response / Q-loss checks and forced-deviation impulse responses (deviation $\rightarrow$ punishment $\rightarrow$ recovery).
- R3.4. “Price Gain” is not a clean collusion measure.** Its denominator ($p^{\text{col}} - p^{\text{comp}}$) moves with γ , so Price Gain can rise even as prices fall. Use direct measures: deviation incentives, punishment behavior, Q-value differences.
- R3.5. Dynamic foundation & convergence reporting.** Convergence guarantees do not apply; ideally give conditions for convergence to fixed points/cycles vs. γ . At minimum report: fixed-point vs. cycle rates, handling of non-converged runs, sensitivity to the convergence criterion, grid size, memory, initial Q , α , β . **Flags an inconsistency:** baseline $\beta = 4 \times 10^{-5}$ (Sec. 5.1) lies outside the robustness grid $[2.8 \times 10^{-7}, 2 \times 10^{-5}]$ (App. C.2).
- R3.6. Narrow the scope.** Results are conditional on two symmetric firms, one algorithm, and selected parameters — and the paper’s own theory says collusion can get *easier* under some conditions. Justify $\gamma \in [0, 3]$ empirically with a mapping to the reference-dependence / loss-aversion literature.
- R3.7. AI-agent implications overreach.** The 2×2 welfare exercise uses two extreme γ values, some differences are tiny (0.722 vs. 0.708), with no sensitivity analysis or uncertainty quantification. Clarify individual vs. collective adoption; separate the positive finding from the normative design recommendation.

R3’s overall verdict: the concerns are not independent; addressing them would require a dynamic theory isolating what reference dependence adds, convergence analysis tied to the learning dynamics, real-world validation, or a much deeper computational analysis — “a different paper rather than a revision.”

6 Cross-cutting themes (how the points cluster)

- A. Identification: reference dependence vs. elasticity rescaling.** (R2.1, R3.1, AE1) — the decisive issue.
- B. Theory–simulation alignment / need for a dynamic model.** (R1.1, R3.1, R3.5, AE1)
- C. Mechanism transparency in the simulations.** (R3.3, R3.4, AE2)
- D. Theory hygiene: Prop. 2 scope; Q-learning overclaiming; equilibrium selection.** (R2.2, R2.3, R3.5)
- E. Scope / calibration of claims.** (R3.2, R3.6, R2.1)
- F. Positioning: AI framing vs. reference-dependence content.** (R1.3, R2.4, R3.7)
- G. Modeling extensions.** loss-aversion asymmetry (R1.2), endogenous references / MSRP (R1.4), n firms (R1.1).
- H. External validation.** human-subject / field data (R1.5, R3.2).

7 Proposed responses and feasibility

Feasibility judged against the existing codebase. Legend: **Easy** (small code/writing), **Moderate** (new experiments/runs), **Heavy** (new theory/section).

Pts	Proposed response	Feasibility / notes
R2.1, R3.1	Add a pure price-sensitivity benchmark $u = \theta - (1 + \gamma)p$ (no r in demand <i>or</i> firm state); overlay on the reference model at equal on-path elasticity. Divergence off-path = the identified reference effect.	Easy : ~15-line <code>demand_type</code> in <code>init.py</code> ; reuse <code>gamma_only</code> pipeline. <i>This directly answers the decisive point.</i>
R2.1, R3.1	Reframe the mechanism around off-path reference gaps : the cheating-deviation and $\bar{\delta}$ terms load on $\gamma(r - p^{\text{cht}}) \neq 0$ even under rational expectations. Make this the headline theoretical distinction from “pure elasticity.”	Easy–Mod : writing + light algebra; no new tooling.
R3.1	Quantify realized $ r - p $ gaps in simulations and decompose the price decline into elasticity vs. reference-gap components.	Easy : cycle prices & reference prices already saved per session.
R1.1, R3.1, R3.5, AE1	Build a dynamic (Markov-perfect) benchmark : with exponential-smoothing reference formation, $r_{t+1} = \lambda r_t + (1 - \lambda) \sum_i D_i p_i$ is a deterministic state transition; solve the duopoly MPE by value iteration on the existing grid. Compare MPE prices to Q-learning across γ .	Heavy (new section) — but this is the <i>single change that best answers the AE</i> . Tractable computationally; a full closed-form MPE is not required.
R3.3, R3.4	Open the black box: save per-session Q ; report best-response / Q -loss of converged strategies and forced-deviation impulse responses (punishment depth, recovery length) vs. γ . Add a direct collusion measure (one-shot deviation gain from the PI tensor) alongside/instead of Price Gain.	Moderate : cycle replay machinery exists; need to persist Q before it is overwritten in <code>run_sessions</code> .
R2.2	Restate Prop. 2 honestly as local/limiting; add a <i>numerical global</i> $\bar{\delta}(\gamma, \mu, c)$ surface.	Easy : extend <code>compute_p_competitive_monopoly</code> with a cheating-price <code>fsolve</code> ; one figure.
R2.3, R3.5	Rewrite convergence discussion: drop Watkins–Dayan, frame as a Markov game with multiplicity, describe Q-learning as an empirical selection device; report selection statistics and criterion sensitivity.	Easy : mostly writing; convergence/cycle stats already collected.
R3.5	Fix the β inconsistency: paper says 4×10^{-5} , code default is 4×10^{-6} (<code>init.py:43</code>), which <i>is</i> inside the robustness grid.	Easy : one-line manuscript fix; confirm the Fig. 3 runs’ value.

Pts	Proposed response	Feasibility / notes
R1.2	Promote loss aversion ($\phi > 1$) toward the baseline rather than an extension.	Easy: already implemented (<code>lossaversion</code> kwarg, Sec. 6.2).
R1.4	Add endogenous reference manipulation (firm-announced MSRP entering reference formation) as an extension.	Moderate: action space k^2 per firm; still tabular. New section.
R1.3, R2.4, R3.7	Reposition: either (a) demote AI to the discussion and lead with “behavioral demand disciplines algorithmic collusion,” or (b) commit to the agent framing via a mixed-population analysis (share of rational $\gamma = 0$ vs. behavioral consumers). Strengthen the Sec. 7 welfare exercise (full γ grid, CIs, individual vs. collective).	(a) Easy ; (b) Moderate (mixture demand). Decision needed with supervisors.
R1.5, R3.2	External validation: lab experiment and/or field data.	Out of scope for the repo. AE says human subjects are secondary <i>if</i> theory is strong, so prioritize the dynamic theory instead.

8 Suggested priorities for the meeting

1. **Decide the core reframing:** anchor the paper on the **off-path reference-gap mechanism** (what reference dependence adds beyond elasticity). This is the make-or-break move.
2. **Commit to a dynamic benchmark** (computational MPE) to align theory with the simulations — the AE’s central ask.
3. **Run the $u = \theta - \beta p$ benchmark** (cheap) as the concrete rebuttal to R2.1.
4. **Add mechanism diagnostics** (impulse responses, deviation incentives) — R3.3/R3.4.
5. **Positioning decision:** lead with reference dependence, or build out the AI-agent analysis?
6. **Note on strategy:** this was a reject (not R&R), and R3 says the fixes “amount to a new paper.” Plan a **fresh submission** (MS resubmission after major restructuring, or an alternative outlet) rather than a point-by-point rebuttal.

9 Detailed proposal: modeling AI consumer agents via LLM measurement

*This is a concrete plan for the extension the referees most wanted (R1.3, R2.4, R3.7): putting AI consumer agents into the **analysis**, not just the motivation. It requires **no change to the pricing/RL architecture**. This version incorporates two rounds of internal design review (identification, elicitation, confounds, and simulation-integration concerns).*

Positioning: what this is and is not

This is an extension, not the fix for the decisive rejection reasons. It does *not* resolve (i) the dynamic/MPE theory gap, (ii) the on-path elasticity-vs-reference identification *in the model*, or (iii) open-the-black-box mechanism diagnostics. Build order: **(1)** dynamic theory + elasticity identification, **(2)** mechanism diagnostics, **(3)** this LLM extension — on top of the foundation, not instead of it.

Honest odds. Before scaling, we run a small pilot (the *gating prototype*, defined step by step below) with five pass/fail checks. We rate the chance that all five come back “pass” at roughly 60–70%; the two most likely failure modes — the *waiting* confound and the *forecast-too-good* degeneracy — are named below with designed mitigations. Either failure would be discovered in a ~\$5 pilot rather than after building the full study, and every outcome, including a null, is informative (see “Why every outcome is a result”).

The problem it solves

The obvious move — add AI agents as a demand segment with a chosen reference sensitivity γ_a — **begs the paper’s own question**. We do not know whether an AI agent is reference-dependent, or its price sensitivity; *setting* γ_a assumes the answer. Instead we **measure** the agent’s revealed behavior, then feed it into the equilibrium model. This matches the cited literature (Ross et al. 2024; Bini et al. 2026): LLMs are “neither fully human nor fully rational,” so behavior must be measured. It also occupies an open niche: Fish et al. (2024) study LLMs as *sellers*; LLM-as-*consumer* measurement fed into an equilibrium pricing model does not yet exist.

Design principle: measure, then simulate (decouple)

An LLM cannot sit inside the Q-learning training loop (cost analysis below: $\sim 10^8$ – 10^9 calls). So:

- **Phase A (LLM, once):** estimate the agent’s choice rule — price coefficient, history-sensitivity parameters, and reference-formation rule — from its choices and forecasts. Cheap; no market loop.
- **Phase B (tabular, as now):** insert the estimated *choice rule and reference rule* into the existing simulation for equilibrium prices, profits, welfare.

Unit of analysis. The estimated parameters are **not** “the LLM’s γ .” They characterize a **shopping policy** induced by a specific *model* $\times$ *system instruction* $\times$ *memory design* $\times$ *task environment*. This is a feature: it makes reference dependence an *engineered design object*, not a fixed trait. γ remains a *knob* in the tabular model; the LLM supplies estimated policies for chosen configurations.

Phase A step by step (the operational pipeline)

For a reader coming to this fresh, here is exactly what we do, in order:

- Step 1. Build the history library from data we already have.** Take the converged sessions of the paper’s baseline 30- γ runs (archived in `paper_results`, `cycle_prices`) and extract representative price paths: steady price ($L=1$), high-low alternation ($L=2$), four-period cycle ($L=4$), Edgeworth cycle ($L=6$). Add two synthetic patterns the runs do not provide — a trend and a volatile path — because they are needed for identification (Step 6 check). Result: ~ 8 history patterns, each a window of recent prices, e.g. [1.50, 1.70, 1.50, 1.70, 1.50].
- Step 2. Set the agent’s memory m** = how many past periods the LLM is shown. $m=0$: no history at all (only today’s two prices — this agent *cannot* form a reference); $m=3$: the last 3 periods; $m=5$: the last 5. Memory is a design knob of the agent and one of our treatments.
- Step 3. Build scenarios.** One scenario = (history pattern, m , today’s prices (p_1, p_2), customer-taste draw, waiting arm, valuation arm). Today’s prices are sometimes *on* the pattern (the expected next leg of the cycle) and sometimes *off* it (a surprise: the cycle says \$1.50 comes next but we show \$1.85) — the surprises create the $r \neq p$ variation that identifies γ . The taste draw is a line such as “your customer mildly prefers Brand A” (randomized symmetrically). The waiting arm is “can wait one period” vs. “must buy today.” Crossing these gives a few hundred to $\sim 1,000$ unique scenarios.
- Step 4. Ask, under three elicitation conditions.** **(A) Natural choice:** history + today’s offers $\rightarrow$ “A, B, or NONE?” — no forecast mentioned. **(B) Forecast-then-choice:** same call, but first “what price do you expect next period?”, then the choice (included only to test whether asking the forecast *changes* behavior). **(C) Independent forecast:** two *separate* call sets — Set 1 sees only the history and forecasts next period’s price (no shopping); Set 2 sees the same history and shops (no forecast question). The averaged Set-1 forecast per history becomes $\hat{r}(h)$, used to analyze Set-2 choices.
- Step 5. Query.** For choice calls, read the token logprobs of “A”/“B”/“NONE” $\rightarrow$ choice probabilities per scenario (one call each, low noise). For forecast calls, read the predicted number.
- Step 6. Estimate.** (i) From Set 1: fit the forecast rule $\hat{r}(h)$ — exponential smoothing (recover $\hat{\lambda}$) or cycle-anticipating? (ii) From the choices: fit the competing choice models (pure price; reference gain/loss using $\hat{r}$; last-price; deal; trend; waiting) by maximum likelihood and compare on *held-out* scenarios. Output per (model, prompt, m , arm) configuration: $\hat{\beta}, \hat{\gamma}_G, \hat{\gamma}_L, \hat{\mu}^A$, the winning specification, and the reference rule $\hat{r}(h)$ for Phase B.

Whose demand curve is it? The constructed-population design

A single LLM is *one* agent; the softness of its token probabilities is **not** a population of heterogeneous consumers. A referee will ask why $P(\text{buy } A)$ from one model’s logprobs should equal a market share D_A . Our answer — which *also* solves the horizontal-differentiation problem — is to **construct the population ourselves**:

- Each call carries a randomly drawn *user taste*: “your customer mildly/strongly prefers brand A” (or B), drawn symmetrically across firms and independently across calls. This is the experimental analog of the paper’s Gumbel taste shock $\varepsilon_{n,i}$: *horizontal* taste — no option is universally better — so it does not smuggle in vertical quality.

- Market demand is then the *average of the agent’s choices over the induced taste distribution*: many delegated shoppers, each serving a different customer — exactly what a deployed shopping agent does at scale.
- What we *measure* is the agent’s **exchange rate between stated taste and dollars**: how large a price premium it will pay to honor the customer’s preference. That exchange rate (together with the taste dispersion we impose) pins the AI segment’s *effective* differentiation parameter $\hat{\mu}^A$ — rather than pretending logprob noise is taste heterogeneity.

This resolves the earlier open question on differentiation: taste lives in the *customer profile we randomize*, not in product quality claims. The remaining open item is only the neutral *wording* of the profile (see prompt principles below).

Phase A — what we estimate (with gain/loss asymmetry)

We do *not* force the paper’s symmetric single- γ form. Estimate a kinked gain–loss response (this also directly answers R1.2):

$$U_i = \alpha_i - \beta p_i + \underbrace{\gamma_G (r_i - p_i)_+}_{\text{gain: below reference}} - \underbrace{\gamma_L (p_i - r_i)_+}_{\text{loss: above reference}} + \varepsilon_i, \quad \text{test } \gamma_L > \gamma_G,$$

recovering $\hat{\alpha}, \hat{\beta}, \hat{\gamma}_G, \hat{\gamma}_L, \hat{\mu}^A$ (with SEs) per agent configuration — *plus* the reference-formation rule $\hat{r}(h)$ from the forecast task (needed for Phase B; see below).

Phase A — experimental design

1. Realistic histories (not constant prices). Markets converge to *cycles* (Fig. 9/10), so the reference is an *anticipated* price, not a flat average. Draw the history library from the actual converged price paths in `paper_results` (`cycle_prices`): fixed point ($L=1$), 2-period ($L=2$), 4-period ($L=4$), Edgeworth ($L=6$) — *plus trend and volatile patterns, which are required for identification* (see the forecast-too-good caveat below).

2. Horizontal differentiation via randomized customer taste. Resolved by the constructed-population design above: each scenario randomizes the customer’s brand taste symmetrically across firms. No quality scores, no vertical ranking.

3. Principal–agent framing: robust but not nudging (wording still to settle). The agent shops *for a user*, not for itself. Prompt principles:

- *Never* mention reference prices, deals, fairness, or comparisons to the past — nothing that leaks the hypothesis.
- *Do not* state a precise valuation (“worth exactly \$2.00”): that reduces the task to arithmetic and can erase the very history-sensitivity we want to measure. Run **two valuation arms**: (i) explicit valuation (clean $\hat{\beta}$; tests whether history *still* moves choices when value is known) and (ii) no valuation (the realistic case, where a reference is the agent’s natural proxy for “is this a good price?”). If $\hat{\gamma}$ appears in (ii) but not (i), that is a *finding*: reference dependence substitutes for missing valuation information, and agent designers can remove it by supplying valuations.
- Candidate skeleton (wording to be finalized): “You are a shopping assistant buying [product] for a customer. [Taste line.] [History block, per arm.] Today’s options: Store A \$ p_1 , Store B \$ p_2 . Reply with A, B, or NONE.”

4. **The LLM never sees “ r ”.** The reference is induced by the displayed history; the LLM only chooses. Move the history *independently* of today’s prices, so at fixed (p_1, p_2) any choice shift is attributable to history, not β . Cycles do this naturally (today = high leg vs. low leg), and **cycles are the $r \neq p$ regime R3.1 said genuine reference dependence lives in**. Read choice probabilities from **token logprobs** where available (one call/scenario, low noise).

5. **The no-waiting arm (kills the waiting confound by design).** Our honest prior is that a capable LLM shown a 1.50/1.70 cycle with today on the high leg will *reason about the cycle and skip* — rational intertemporal substitution, **not** reference-dependent utility (the paper’s consumers cannot time purchases). So include an arm where waiting is impossible: “*the purchase must be completed today; the customer needs the product tonight.*” If the history effect *survives* when waiting is ruled out, it cannot be waiting. Comparing the with- and without-waiting arms separates the two mechanisms — and the size of the waiting response is itself a reportable behavior of AI shoppers.

6. **Identification caveat: a forecaster that is too good destroys identification.** γ_G and γ_L are identified only if today’s price falls both *above and below* the agent’s own reference $\hat{r}$. If the LLM forecasts a clean cycle perfectly, then $\hat{r}$ equals today’s leg, $r - p \approx 0$ everywhere, and *nothing* is identified — ironically, the better the agent’s forecasting, the worse our identification. Mitigations: (i) include trend, volatile, and phase-ambiguous histories where forecasts are necessarily imperfect; (ii) include out-of-pattern “surprise” prices (today’s price deviates from the learned cycle — exactly the off-path deviations that matter in the theory); (iii) verify ex post that the realized $|\hat{r} - p|$ distribution has support on both sides of zero.

Separate the forecast from the choice (avoid manufacturing the effect)

Asking “what do you expect next?” immediately before “buy?” can **create** the reference dependence we are trying to detect (salience/demand effect). Use three elicitation conditions:

- (A) **Natural choice** — history + offers, choose. No forecast asked.
- (B) **Forecast-then-choice** — forecast, then choose (same call).
- (C) **Independent forecast** — a *separate* set of calls forecasts prices; choice calls use identical histories *without* being asked to forecast. Use the averaged independent forecast as $\hat{r}$.

Condition (C) is the defensible one for estimation. If the history effect appears only in (B), it is elicitation-induced, not a genuine shopping behavior. The (C) forecasts serve double duty: they are also the data from which we fit the agent’s *reference-formation rule* $\hat{r}(h)$ for Phase B.

Confounds and model comparison (the go/no-go)

A nonzero history effect need not be reference dependence. It could be strategic waiting, trend extrapolation, a “good deal” heuristic, a quality signal, or manipulation-suspicion. **The waiting confound is existential** — hence the dedicated no-waiting arm (design item 5) *plus* model comparison. Estimate competing specifications and select on **held-out** predictive accuracy (train/test on disjoint histories, levels, cycle patterns, products):

- pure price: $U_i = \alpha_i - \beta p_i$;
- expected-price reference (gain/loss): $+\gamma_G(\hat{r}_i - p_i)_+ - \gamma_L(p_i - \hat{r}_i)_+$;
- last-price reference: $+\gamma_L(p_{i,t-1} - p_i)$;
- range / “deal”: $+\gamma_{\min}(p_{\min,h} - p_i) + \gamma_{\max}(p_{\max,h} - p_i)$;

- trend: $+\tau \cdot \text{trend}_h$;
- **waiting value** in the outside option: $U_0 = \alpha_0 + \omega(\hat{r} - p_{\min,t})$ — estimated in the *with-waiting* arm; should shrink toward zero in the no-waiting arm.

The reference model earns its place only if it beats these out of sample.

Information-set arms (memory as the design lever)

Memory is what makes a reference *possible*, so the information set is the treatment:

- (a) **No-history** — today’s prices only. **Baseline price-response function; with no history variation** γ is *not identified* here — a benchmark, not a “ $\gamma = 0$ ” validity test.
- (b) **Price-history** — + observed price path. Observation-based reference.
- (c) **Transaction-history** — + “last period you bought from B at \$1.60.” Paid-price / status-quo reference.

The across-arm comparison **is** a core contribution: *giving a consumer AI agent memory can manufacture reference dependence, and which memory determines which reference forms* (arm (b) $\approx$ common reference; arm (c) $\approx$ transaction/firm-specific, Section 6.3.1).

Phase B — market simulation with AI adoption

What Phase B actually needs (previously under-specified). The simulation needs *two* objects for the AI segment, not one:

1. the **choice rule**: $(\hat{\alpha}, \hat{\beta}, \hat{\gamma}_G, \hat{\gamma}_L, \hat{\mu}^A)$ — the full vector, not a $\hat{\gamma}/\hat{\beta}$ ratio;
2. the **reference-formation rule** $\hat{r}(h)$: what reference the AI segment carries *as the simulated market evolves*. We obtain it by fitting the Condition-(C) forecasts to candidate rules (e.g. exponential smoothing $\Rightarrow$ an estimated $\hat{\lambda}^A$; or a cycle-anticipating rule), and implement it with the code’s existing `ref_prediction` machinery.

Implementation v1 (keeps the code tractable). If humans and AI agents carry *different* reference processes, the firms’ state must track two reference variables and the state space multiplies by k (15 $\times$). Version 1 therefore uses a **shared common reference process** for both segments, with segments differing in $(\beta, \gamma_G, \gamma_L)$; changes are contained to `demand()/foc()` and the profit tensor in `init.py`, with no state-space blowup. Separate reference processes are a v2 extension, to be run only if the fitted $\hat{r}(h)$ rules differ materially between segments.

Adoption, not replacement. With AI-agent share $\eta \in [0, 1]$,

$$D_i(\eta) = (1 - \eta) D_i^H + \eta D_i^A,$$

sweep η (the mixture idea, now *earned* because D_i^A is measured, not assumed). Vary η , memory design, agent objective, and model configuration; report prices, profits, consumer surplus, purchase delay, and collusion diagnostics. Keep the Fig. 3 γ -axis placement only as an intuition visual, not the structural mapping. The η -sweep is the direct answer to **R3.7** (individual vs. collective adoption). **Welfare bookkeeping**: evaluate the AI segment’s surplus under both the bias and preference conventions of Section 7, so the 2×2 logic carries over with measured inputs.

Phase C — dynamic validation (cheap)

Do not retrain with an LLM in the loop. Instead: train seller algorithms against the *estimated* AI demand; freeze the learned seller policies; replay a manageable set of market states; ask the *actual* LLM to choose in those states; compare to the fitted demand. This tests whether the static estimate survives in dynamically generated states ($\sim 10^3$ – 10^4 calls; see cost table).

Why every outcome is a result

- $\hat{\gamma} \approx 0$: agentic commerce *removes* the consumer-side discipline human reference dependence provided — a policy warning.
- $\hat{\gamma} > 0$: the safeguard survives delegation — reassuring, and quantified.
- Effects differ across memory arms: the **design-lever result** — agent memory architecture is a competition-policy variable.
- Large waiting response: a measured behavior of AI shoppers with its own pricing implications.
- Prompt-unstable $\hat{\gamma}$: evidence that deployed agent shopping policies are fragile — itself relevant to the consumer-agent design debate.

Cost analysis

Design	LLM calls	Cost	Verdict
Phase A (measure), few models	~ 3 – 10 k	$\sim \$10$ – 50	do this
Phase C dynamic validation (freeze sellers, replay)	$\sim 10^3$ – 10^4	$\sim \$20$ – 100	feasible check
In-loop <i>training</i> , 1 γ (LLM’s own), 200 sessions	$\sim 10^8$	$\\$100$–$500$k	infeasible
In-loop training, full sweep (200 sess $\times 30 \gamma$)	$\sim 3 \times 10^9$	$\\$3$–$15$M	infeasible

Drivers: $\sim 5 \times 10^5$ periods to convergence $\times$ sessions, one LLM call/period. **Because an LLM has its own γ , in-loop is one γ not 30 — but periods $\times$ sessions still makes training infeasible.** Phase-A traps: avoid reasoning models with long chains-of-thought (force terse output); prefer logprob-capable models (sampling is the expensive part). Sample size is set by the number of *unique scenarios*, not total calls — design the scenario matrix first and check precision at the scenario level.

Gating prototype — do this before scaling

The prototype is simply **Steps 1–6 at miniature scale**: 4 history patterns (incl. one trend, one volatile) $\times$ 5 price configs $\times$ 2 customer-taste draws $\times m \in \{0, 3\} \times \{\text{waiting, no-waiting}\}$, elicitation conditions (A) and (C), one logprob-capable model — a few hundred calls, roughly \$5, a few days of work. “**Surviving**” the prototype means all five of these checks come back *pass*:

Q1. The fit *separately* recovers β and the history effect (they are not collinear).

Q2. Adding history improves *held-out* choice prediction beyond current prices alone.

- Q3.** The history effect *survives the must-buy-today arm* — so it is genuinely $r - p$, not waiting/trend.
- Q4.** The realized $\hat{r} - p$ gaps have support on *both* sides of zero (the surprise prices worked; identification not destroyed by good forecasting).
- Q5.** Plugging the estimates into the simulation moves market prices *non-trivially* in Phase B.

If any answer is “no,” stop — do not scale. Honest risks: the waiting confound may dominate (our modal-failure candidate, caught by Q3), or forecasts may be too accurate to leave identifying variation (caught by Q4), or $\hat{\gamma}$ may be too prompt-unstable to carry signal. The prototype is a genuine go/no-go, and even a “no” is a publishable observation about AI shopper behavior.

Which referee points this addresses (and which it does not)

Points	How
R1.3, R2.4	AI agents become market participants with <i>measured</i> behavior and an adoption sweep; the AI framing becomes load-bearing.
R3.7	η -sweep separates individual choice from collective/market effect; replaces the stylized 2×2 with measured behavior and CIs.
R1.2	Gain/loss asymmetry ($\gamma_G \neq \gamma_L$) estimated directly.
R3.1, R2.1 (<i>partial</i>)	Estimates γ and β <i>separately</i> in the $r \neq p$ regime, giving γ empirical content distinct from elasticity. Does not fix the on-path identification in the theory model.
Sec. 6.3	Forecast elicitation locates the agent between exp-smoothing and Q-learning reference formation — and supplies the $\hat{r}(h)$ rule Phase B needs.
Not dressed	ad- Dynamic/MPE theory; on-path elasticity identification in the model; open-the-black-box mechanism diagnostics. These require the foundation work, not this extension.

The reframed contribution (one line)

We experimentally manipulate the memory and objectives of AI shopping agents, estimate how these designs shape their price expectations and purchasing behavior, and quantify how consumer-side AI adoption changes the pricing strategies learned by seller algorithms. That is a legitimate AI-agent contribution — and it avoids the awkward “give agents human biases” recommendation by framing reference dependence as an engineered design choice.